Lecture 6 (25/05/21) Ch 3.1

Vector Spaces and Subspaces

Vector Space

- Bunch of vectors or space of vectors

for example, R^2 is a vector space,

where R^2 is all 2 dimensional real vectors = xy plane

$\begin{bmatrix} 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \pi \\ e \end{bmatrix}$

Characteristics of vector space

* If we add two vectors space and multiply any vector with a scalar, we should be inside the vector space after that. So the result should stay inside the space.

The linear combination of any two vectors in that space should result in that space as well.

The most crucial element of vector space can't be removed.

Two vectors

$$\begin{tikzcd} & u \arrow[dl] & & v \arrow[dr] & \\ cu \arrow[dr] & & & & dv \arrow[dl] \\ & & cu+dv & & \end{tikzcd}$$

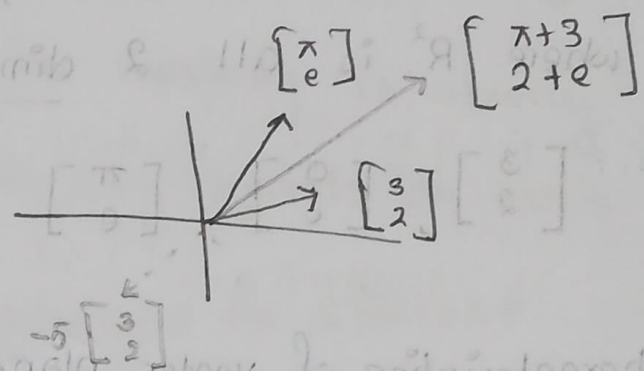
scalar multiplication

vector addition

$\underbrace{\qquad\qquad\qquad}_{\text{Linear combination}}$

<u>Example:</u>

$$\begin{bmatrix} 3 \\ 2 \end{bmatrix} \begin{bmatrix} \pi \\ e \end{bmatrix}$$



$\begin{bmatrix} \pi+3 \\ e+2 \end{bmatrix}$ will be in vector space.

- R^2 fulfils the condition of being vector space.
- If we take out $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ point from any vector space, then can we call it a vector space?

No, if we multiply a vector with scalar $c=0$, the result will be at the origin, if we remove 0 from vector space, the linear combination/scaling won't stay in the vector space after that.

To be vectorspace

1. The <u>linear combination</u> of any two vectors from the vector space should stay inside the space.

2. The <u>origin</u> must be <u>inside</u> the vector space.

* To create a real vectorspace we should be able to produce vectors by addition and multiplication. That follows certain laws as commutative, distributive laws.

$$v + w = w + v \quad \text{Commutative Law}$$

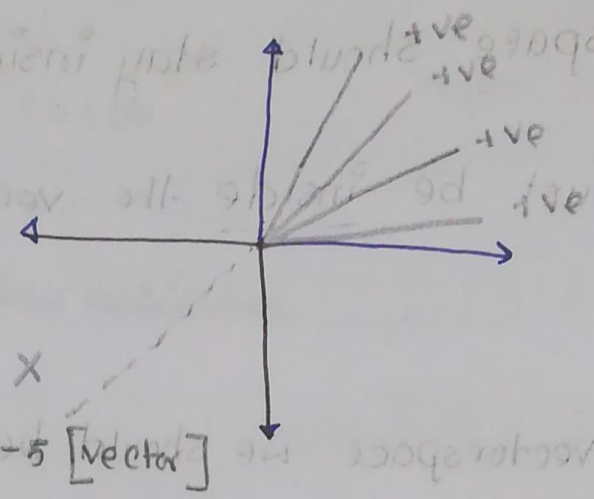
$$c(v+w) = c(v+w) \quad \text{Distributive Law}$$

* we can produce infinite number of vectors using the above laws in the vector space.

* Vector space is a very strong concept in linear algebra.

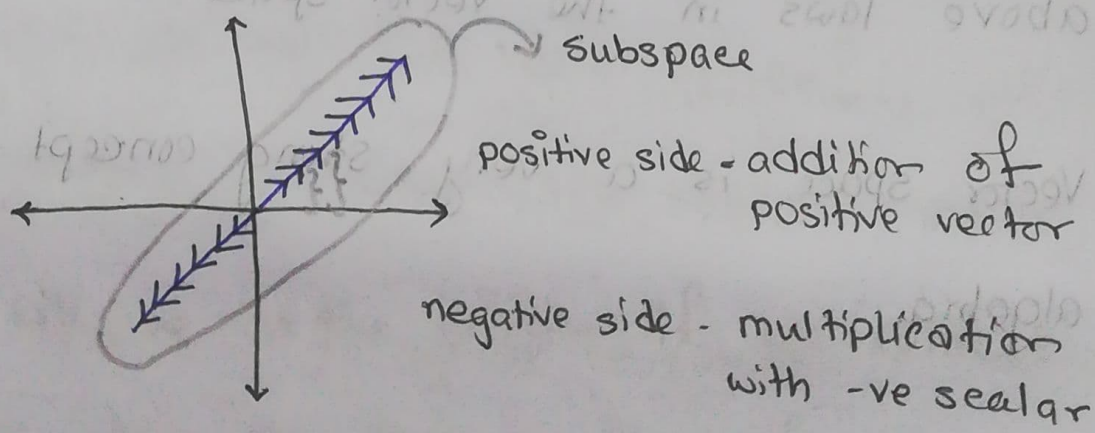
Example of not a vector space

(vectorspace is the first quadrant in this case)



This violates the first rule (3rd quadrant) as it lands outside vector space (1st quadrant) after multiplying by -5 (scalar).

Tell me a vector space that is inside R^2 and goes by multiplication and addition.



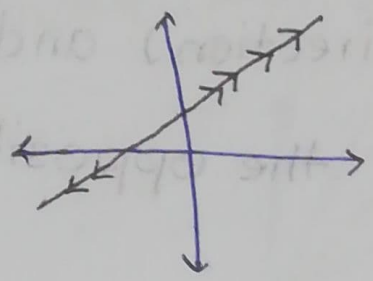
By using our imagination, if we think about any vector, and consider it's positive sealars (that goes in the same direction) and its negative vector (which goes in the opposite vector) towards infinity

Now if we take any two vectors (in the positive direction) and add them, then it will stay in the positive side (of the vectorspace). Again if we take any vector from the positive part and multiply it with a negative sealar, we'll stay at the opposite direction

So, we can see that, the vectorspace inside R^2 is closed by addition and multiplication inside its own space. Thus, it can be called a vectorspace. But as it is a vector space inside another vector space, this is called a subspace.

Can any line be called a vector subspace?

For example,

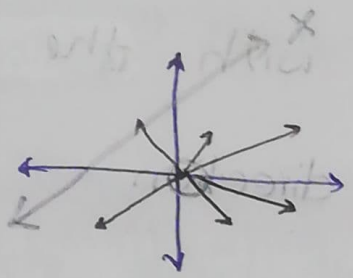


Is this line a vector subspace?

- No. since it violates the second rule.

(It doesn't contain the origin)

Why having the origin is important?



"A line could be a subspace if it goes through origin."

Lets consider there are many vectors in the vectorspace, if we add two vectors, they might cancel each other out and we can get the origin. But the straight line isn't going through the origin, so the straight line isn't inside vector space.

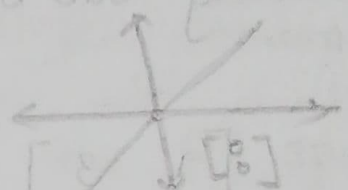
R^3 is a vector space that contains all 3 dimensional real vectors such as $\begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$.

R^n is a vector space that contains all n dimensional real vectors, such as $\begin{bmatrix} x_0 \\ \vdots \\ x_n \end{bmatrix}$

What are all possible sub-spaces of R^2 ?

① R^2 itself (every set is a subset of itself.

② A line through the origin



③ Zero vector / origin $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$

(Any linear combination of zero vector will stay inside 0 vector / origin) - [2 conditions satisfied]

The conditions to be a vector subspace is same as the conditions to be a vector space.

To be a <u>vector subspace</u>

① The linear combination between any two vectors should be inside the vector subspace

② The origin should be inside vector subspace

What are all the possible subspaces of R^3?

① The linear combination of any two vectors should lie inside the vector subspace

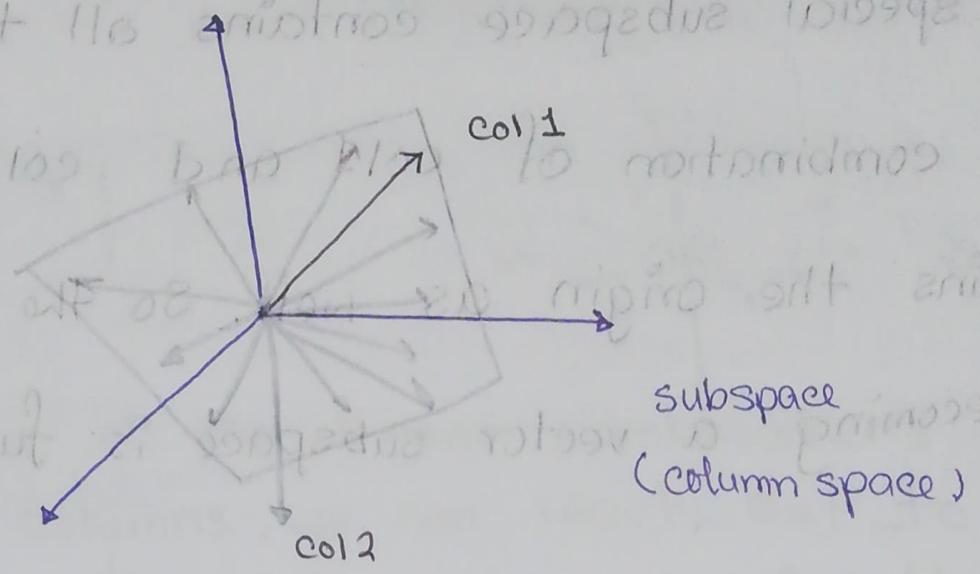
② The planes passing through the origin

Creating sub spaces out out matrices.

$$A = \begin{bmatrix} 1 & 3 \\ 2 & -3 \\ 4 & 1 \end{bmatrix}$$ (These are the columns in R^3)

In order to call it a vector subspace, we need to be able to take all the linear combinations of the columns to call it a vector subspace.

All the linear combinations which creates the subspace are called the <u>column space</u>.



We should be able to take the linear combination of the two columns. So, if we take their linear combination, we'll get a plane like structure in the R^3 space which is a plane that goes through the origin. And the plane is a subspace as it fulfils all the conditions

* How do I include the origin?

- We take the scalars as 0

$0[\text{col } 1] + 0[\text{col } 2]$ which includes the origin.

This special subspace contains all the vector's linear combination of col1 and col2 and it contains the origin as well. So the two conditions of becoming a vector subspace is fulfilled.

And as this subspace is made up of the columns of the given matrix, this subspace is called the columnspace of that particular matrix.

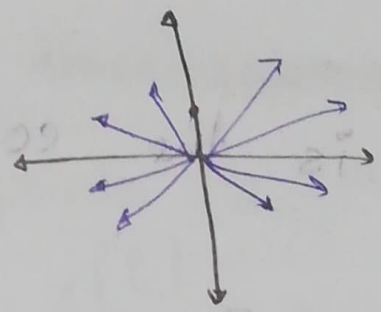
Column Space:

The area or the vectorspace that is covered by the columns of any matrix.

- A plane that goes through the origin in R^3.

Can you describe the column space of following matrices?

① $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$



If we take all the linear combinations of these two columns, we can reach out to any point of R^2. So, every vector is the combination of the columns of I_2. we can reach any point.

Lets consider a point,

$\begin{bmatrix} -2000 \\ 1500 \end{bmatrix}$ can we reach this point by the help of colums of I_2?

- Yes. How? $-2000 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1500 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Similarly we can reach out to any point by linear combination of the two colums.

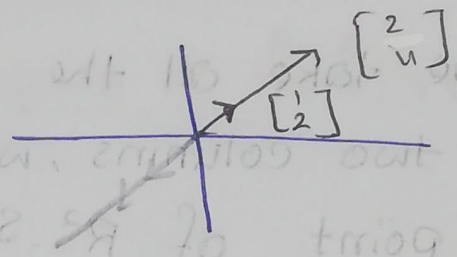
"We can reach out to any points in R^2 by the columns of IDENTITY MATRIX"

"I CAN COVER THE WHOLE VECTORSPACE"

" R^2 is a subspace of I_2 "

" R^2 is the column space of I_2 "

② $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$



Can we reach out to any point with the help of these two vectors?

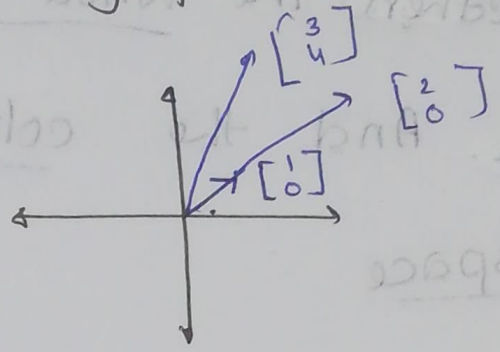
We can't reach 2nd and 4th quadrant

- By taking all the possible linear combinations we can't reach to every point / can't cover the whole R^2 region. We can only cover a line

So its column space is a line through the origin in R^2.

③ $B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 4 \end{bmatrix}$

By the help of these three colums, we can reach out to any points by linear combination



Here every b is attainable, by the help of these three colums. $(Ax = b)$

$\therefore$ The column space is whole R^2

* $\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$ is similar to $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$

* From these three examples we can guess that n-dimensional matrix having independent columns can fill the whole vector space.

And the columnspace becomes a vectorspace

* n columns are enough to fill up a whole vector space of n dimensions.